Jan-Paul Vincent Ramos-Dávila

Personal Data

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Experiences

2026 - **Network Systems Software Engineer, Hewlett-Packard Enterprise**
As a Network Test Engineer (II), I design and execute automated system test plans for HPE ArubaOS-CX campus and branch switches within the EDSW team, validating enterprise Layer 2/Layer 3 protocols, Dynamic Segmentation, and high-density traffic performance.
Develop scalable Python and PyTest automation frameworks integrated into CI/CD pipelines to isolate firmware defects, capture packet traces, and drive critical software updates across Aruba switching platforms.
Supervisor: Solution Test and Auto ENG (Sanjeev Kumar)

2024 - 2025 **Research Intern, NASA Langley Formal Methods**
Mechanized proofs that model correct behaviors of a Software Defined Delay-Tolerant Network's Match-Action pipeline for NASA's Interplanetary Overlay Network framework.
Developed a formally verified Network Calculus IR in Rocq. Wrote an interpreter for a subset of P4 to target the IR.
Advisor: Dr. Alwyn Goodloe

2022 - 2024 **Research Intern, Carnegie Mellon University S3D**
Core contributor on the early development of the Gradual Verification framework. Empirically evaluated the soundness of Gradual C_0, and provided formal proofs of completeness between the dynamic and static verifiers.
Explored the application of Gradual Verification to smart contracts on the Algorand and Ethereum blockchain platforms and developed a prototype for Gradually Verified Teal.
Advisor: Dr. Jonathan Aldrich

2022 - 2024 **Research Assistant, Cornell University CAPRA**
Worked on a symbolic execution tool for verifying parallelism in Calyx.
Advisor: Dr. Adrian Sampson

Education

2025 - On Pause **Boston University**
Doctor of Philosophy in Computer Science
Interests: Programming Languages, Type Theory, Formal Methods
Advisors: Dr. Ankush Das, Dr. Marco Gaboardi

2021 - 2025 **Cornell University**
Bachelor's of Arts in Philosophy
Interests: Foundations of Mathematics and Logics, Analytics
Advisors: Dr. Adrian Sampson (CompSci), Dr. Harold Hodes (Phil)

Summers UC/EasyUC'25, SPLV'25, OPLSS'24, AFP'23

Publications

2025 Jenna DiVincenzo, Ian McCormack, Conrad Zimmerman, Hemant Gouni, Jacob Gorenburg, **Jan-Paul Ramos-Dávila**, Mona Zhang, Joshua Sunshine, Éric Tanter, Jonathan Aldrich. *"Gradual C0: Symbolic Execution for Gradual Verification"*, In **TOPLAS, 46(4), Article No.: 14 P.1-57** and **POPL 2025**

2023 **Jan-Paul Ramos-Dávila**.*"Evaluation Soundness of a Gradual Verifier with Property Based Testing"*, In **Cornell Undergraduate Research Journal, 2(1), P.17-27** and **POPL 2023 Student Research Competition**.

Presentations

2025 *"Sound Default-Typed Scheme"*, In **Scheme and Functional Programming Workshop**.

"Formal Verification of a Software Defined Delay-Tolerant Network", In **IEEE Workshop on Optimizing Interplanetary Communication Through Network Autonomy** and **The Eleventh International Workshop on Coq for Programming Languages**.

2024 *"Gradual Verification of Smart Contracts"*, In **Workshop on Principles of Secure Compilation** and **POPL 2024 Student Research Competition**.

2023 *"Optimization of a Gradual Verifier: Lazy evaluation of Iso-recursive Predicates as Equi-recursive at Runtime"*, In **The Midwest PL Summit 2023** and **POPL 2023 Student Research Competition**.

Teaching

Teaching Assistant

2025 **CS 4/5111 Practicum in Operating Systems**
Ran coding workshops with hands-on demos building and debugging C applications while teaching the EGOS operating system.
Cornell University

2024 **CS 4114 Systems Programming**
Graded assignments and ran coding workshops with hands-on demos building and debugging C++/Linux applications.
Cornell University

CS 4/5110 Programming Languages and Logics
Examination czar in charge of the infrastructure of midterms, graded students' assignments, and held weekly office hours.
Cornell University

Awards

2025	**Graduate Fellowship**, Boston University
	Scholarship, SPLV Summer School at The University of Edinburgh
2024	**Scholarship**, Verification Mentoring Workshop at CAV
2023	**Intern Fellowship**, Amazon Summer Undergrad Research Experience at CMU REUSE
	Third Place Winner, Student Research Competition at ACM SIGPLAN POPL
2022	**Scholarship**, PLMW at ACM SIGPLAN PLDI
2020/1	**Finalist**, Mathematics at Regeneron International Science and Engineering Fair

Academic Service

2027	**Video Co-Chair**, POPL CDMX MX
2026	**Video Co-Chair**, PLDI Boulder CO
	Artifact Evaluation Program Committee, ETAPS Turin IT
	Student Volunteer, POPL Rennes FR
2025	**Video Co-Chair**, ICFP/SPLASH Singapore
	Video Co-Chair, PLDI Seoul KR
	Video Co-Chair, POPL Denver CO
2024	**Virtualization Chair**, ICFP Milan IT
	Virtualization Chair, PLDI Copenhagen DK
2024	**Video Co-Chair**, POPL London UK
2023	**Video Co-Chair**, SPLASH Cascais PT
	Student Volunteer, ICFP Seattle WA

Certificates

In Progress	**(HPE7-A07) HPE Aruba Networking Certified Expert, Campus Access Mobility**
In Progress	**(HPE7-A06) HPE Aruba Networking Certified Expert, Campus Access Switching**
In Progress	**(HPE7-A02) HPE Aruba Networking Certified Network Security Professional**
In Progress	**(HPE7-A01) HPE Aruba Networking Certified Professional, Campus Access**

Skills

English	Native
Spanish	Native
Tools	Unix, Git, Bash, Neovim, Docker, HTML/CSS
PLangs	LaTeX, Rocq, OCaml, Scala, Python, Haskell, JS/TS, Java, Go, C/C++, Rust, P4